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"Adaptive Thresholding" Method to Select Prominent Objects by Converting the Image to Black and White Format

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***Abstract*— Image segmentation is a central task in computer vision, pattern recognition, and digital image analysis, as it enables the extraction of meaningful objects from complex backgrounds. However, real-world scenarios often involve non-uniform illumination, noise, and low contrast, which make traditional global thresholding methods, such as Otsu's algorithm, unreliable. Adaptive Thresholding has emerged as an effective solution by assigning a local threshold to each pixel based on neighborhood statistics. This localized approach enhances object detection in images affected by uneven brightness or distortions. The method has proven useful across diverse domains, including document image analysis, medical diagnostics such as ultrasound and MRI, biometric systems, and intelligent video surveillance. This paper explores Adaptive Thresholding as a strategy for converting grayscale images into binary representations suitable for salient object detection. A theoretical overview of global and adaptive thresholding is provided, followed by mathematical formulations of mean-based and Gaussian-based approaches. Comparative results show that adaptive methods significantly outperform global techniques under challenging lighting conditions. Implementation strategies, algorithm design, and MATLAB experiments are presented, demonstrating improved segmentation accuracy and robustness. Furthermore, practical case studies confirm the effectiveness of adaptive methods in various applications, highlighting their relevance for modern image processing and motivating further research into optimized implementations.**

Keywords—Adaptive Thresholding, Image Binarization, Object Detection, Local Thresholding, MATLAB Implementation, Computer Vision, Image Segmentation.

I. Introduction

A. Background and Motivation

The rapid advancement of computer vision and machine learning technologies has placed digital image processing at the forefront of scientific research and industrial applications. The ability to extract, classify, and analyze meaningful information from raw image data is crucial in fields such as medical imaging, security and surveillance, industrial automation, robotics, and intelligent transportation systems [1]. Among the fundamental tasks in this domain is the conversion of grayscale images into binary representations that highlight significant structures while discarding irrelevant details.

Thresholding is the simplest and most widely used image segmentation technique. It aims to partition an image into foreground (object) and background (non-object) regions by applying a threshold value. Classical methods such as Otsu's algorithm, entropy-based thresholding, and histogram-based approaches assume that a single threshold is sufficient for the entire image [2]. While effective for uniform illumination, such global techniques frequently produce errors when applied to images with shadows, reflections, or spatially varying intensity distributions.

For instance, in medical ultrasound images, different tissues may exhibit local intensity variations caused by equipment noise or patient movement. Similarly, in document image recognition, uneven lighting or paper texture can prevent the proper binarization of text characters [3]. These problems motivate the exploration of more flexible methods, such as Adaptive Thresholding, which can adjust dynamically to local brightness and contrast.

B. Literature Review

Numerous researchers have investigated thresholding techniques over the past decades. Gonzalez and Woods provided a comprehensive overview of classical segmentation methods, emphasizing the limitations of global strategies. Otsu proposed a histogram-based global thresholding method that remains widely cited but is less effective under variable illumination.

To overcome these shortcomings, local thresholding methods were developed. Sauvola and Pietikäinen introduced an adaptive binarization method specifically for document images, which calculates thresholds using local mean and standard deviation. Bradley and Roth further optimized this process using integral images, reducing computational complexity and enabling real-time applications [4].

In the field of medical imaging, adaptive approaches have proven critical. For example, Shapiro and Stockman demonstrated that adaptive techniques outperform global ones when analyzing X-ray and ultrasound images, where intensity distribution is highly non-uniform [5]. Recent studies in biometric authentication and automated surveillance also highlight the significance of adaptive strategies for robust object detection under uncontrolled environments [6].

Thus, the literature strongly indicates that Adaptive Thresholding is not only a theoretical solution but also a practical necessity for real-world object detection tasks.

C. Contribution of the Paper

The main contributions of this paper are as follows:

- A structured overview of thresholding methods, focusing on the transition from global to adaptive techniques.
- A detailed mathematical formulation of Adaptive Thresholding based on local mean and Gaussian-weighted filtering.
- Comparative analysis of adaptive methods with global techniques, demonstrating improved performance in images with non-uniform lighting.
- Application-oriented discussion, particularly in medical diagnostics, document recognition, and surveillance systems.

By addressing these points, the paper contributes to the growing body of literature on robust image segmentation and object detection.

D. Structure of the Paper

The remainder of this paper is organized as follows: Section II formally states the segmentation problem and its challenges. Section III discusses the adaptive thresholding algorithms in detail, including mathematical formulations, algorithmic workflows, and implementation strategies. Finally, the conclusion summarizes the findings and highlights future research directions.

II. Statement of the Problem

A. General Problem Formulation

The fundamental goal of image segmentation is to separate meaningful structures (foreground) from irrelevant background regions. This process typically begins with converting a grayscale image into a binary representation. Let the input image be denoted as:

$$I(x,y):\Omega \subset \mathbb{R}^2 \rightarrow [0,255] \quad (1)$$

where $I(x,y)$ represents the intensity value of a pixel at coordinates (x,y), and Ω denotes the spatial domain of the image.

The objective of thresholding is to define a mapping function $B(x,y)$ that produces a binary image:

$$B(x,y)=\begin{cases}1, & I(x,y)\geq T\\ 0, & I(x,y)<T\end{cases} \quad (2)$$

where T is the threshold value.

If is constant for the entire image, the method is called global thresholding. However, the challenge arises when illumination varies across different regions of the image. In such cases, a single threshold value cannot separate foreground and background accurately.

B. Limitations of Global Thresholding

Global methods such as Otsu's algorithm compute a single threshold value by maximizing the between-class variance of the histogram. Formally, the image histogram is modeled as two classes C0 and C1 separated at threshold .

The within-class variance is:

$$\sigma_w^2(t)=\omega_0(t)\sigma_0^2(t)+\omega_1(t)\sigma_1^2(t) \quad (3)$$

where:

- $\omega_0(t)$ and $\omega_1(t)$ are the probabilities of the two classes,
- $\sigma_w^2(t)$ and $\sigma_1^2(t)$ are the variances of the classes.

The optimal global threshold is chosen as:

$$T^* = \arg\min_t \sigma_w^2(t). \quad (4)$$

Although Otsu's method works well for images with bimodal histograms, it fails in the presence of uneven lighting, shadow effects, or local contrast variations [7]. In such cases, foreground objects may be merged with the background, or background noise may be falsely identified as an object.

C. Adaptive Thresholding Concept

To overcome these limitations, Adaptive Thresholding assigns a unique threshold value to each pixel (x,y) based on the local neighborhood. Instead of a global constant T, the threshold becomes a function:

$$T(x,y)=f\big(N(x,y)\big) \quad (5)$$

where $N(x,y)$ denotes the local neighborhood (e.g., a square window of size $w\times w$) around the pixel.

The binary mapping becomes:

$$B(x,y)=\begin{cases}1, & I(x,y)\geq T(x,y)\\ 0, & I(x,y)<T(x,y)\end{cases} \quad (6)$$

This local adaptation allows the algorithm to adjust to different brightness levels in various regions of the image [8].

D. Mathematical Models of Adaptive Thresholding

1. *Mean-based Adaptive Thresholding*
   The simplest form calculates the mean of the local neighborhood [9]:

$$T(x,y)=\frac{1}{|N(x,y)|}\sum_{(i,j)\in N(x,y)} I(i,j)-C \quad (7)$$

where C is a constant that adjusts sensitivity.

2. *Gaussian Adaptive Thresholding*
A weighted average is computed using a Gaussian kernel $G(i,j)$:

$$T(x,y)=\frac{\sum_{(i,j)\in N(x,y)} I(i,j)\cdot G(i,j)}{\sum_{(i,j)\in N(x,y)} G(i,j)}-C \qquad (8)$$

Here, closer pixels contribute more strongly than distant ones, making this approach more robust to noise.

3. *Niblack's Method*
Another adaptive method considers both mean and standard deviation of the neighborhood:

$$T(x,y)=\mu(x,y)+k\cdot\sigma(x,y) \qquad (9)$$

where $\mu(x,y)$ and $\sigma(x,y)$ are the local mean and standard deviation, and k is a tunable constant.

4. *Sauvola's Method*
An improved version of Niblack's method:

$$T(x,y)=\mu(x,y)\left[1+k\left(\frac{\sigma(x,y)}{R}-1\right)\right] \qquad (10)$$

where R is the dynamic range of standard deviation (typically R=128R), and k∈[0.2,0.5].

E. Practical Problem Definition

Given the theoretical formulations, the practical research problem can be stated as follows:

- **Input**: A grayscale image with non-uniform illumination and noise.
- Goal: Convert it to a binary image such that salient objects (e.g., text in documents, organs in medical images, vehicles in surveillance videos) are accurately extracted.
- **Constraint**: Global methods fail due to illumination variation; therefore, the threshold must adapt locally.
- Solution Direction: Implement Adaptive Thresholding methods (Mean, Gaussian, Niblack, Sauvola) and compare them with global threshold ing approaches.

Justification of the chosen methods.

In this study, the selection of binarization methods was based on their theoretical representativeness and practical relevance across different illumination conditions.

Otsu's global thresholding was chosen as a baseline method because it remains the most widely cited and computationally efficient global approach. It provides a useful reference point for quantifying the advantages of adaptive methods.

Mean and Gaussian Adaptive Thresholding represent two principal local-statistical approaches. The mean-based variant emphasizes computational simplicity and real-time implementation feasibility, while the Gaussian-weighted version improves robustness against random noise by giving higher weight to central pixels in the neighborhood window.

Niblack and Sauvola methods extend the concept by incorporating both the local mean and standard deviation, thereby accounting for local contrast variations that are typical in degraded document and medical ultrasound images.

These four adaptive techniques together form a comprehensive evaluation framework covering both classical (statistical) and enhanced (variance-based) local thresholding strategies. Their inclusion allows for balanced comparison in terms of accuracy, stability, and computational complexity under diverse imaging conditions such as non-uniform illumination, noise interference, and texture variations.

III. METHODS FOR SOLVING THE PROBLEM.

1. General Algorithmic Framework

The Adaptive Thresholding process can be expressed as a series of well-defined steps. Unlike global thresholding, where a single threshold is applied to the entire image, adaptive methods compute a threshold value for each pixel based on its local neighborhood. The general workflow can be summarized as follows [10]:

Algorithm 1: Adaptive Thresholding

1. **Input**: Grayscale image $I(x,y)$.
2. **Preprocessing**: Optionally apply smoothing (Gaussian filter) to reduce noise.
3. **Neighborhood selection**: For each pixel (x,y), define a window $N(x,y)$ of size $w\times w$.
4. **Threshold calculation**: Compute $T(x,y)$ using local statistics (mean, Gaussian, or standard deviation).
5. **Binarization rule**:

Input Image $\rightarrow$ Local Window Selection $\rightarrow$

Threshold Computation $\rightarrow$ Binary Output

Output: Binary image $B(x,y)$.

2. Step-by-Step Mathematical Model

Gaussian-weighted approaches provide higher weight to closer pixels, reducing sensitivity to random noise [11]. Similarly, statistical models such as Niblack's method extend the use of mean and variance, offering robust segmentation in challenging environments.

a) 2.1 Local Mean-Based Thresholding

For each pixel, compute the mean intensity in the neighborhood:

$$T(x,y) = \frac{1}{|N(x,y)|} \sum_{(i,j)\in N(x,y)} I(i,j) - C \quad (11)$$

where C is a constant offset.

b) 2.2 Gaussian-Weighted Thresholding

A Gaussian kernel $G(i,j)$ provides higher weight to closer pixels:

$$T(x,y) = \frac{\sum_{(i,j)\in N(x,y)} I(i,j)\cdot G(i,j)}{\sum_{(i,j)\in N(x,y)} G(i,j)} - C \quad (12)$$

This reduces sensitivity to random noise.

2.3 Standard Deviation-Based Thresholding (Niblack [7])

$$T(x,y) = \mu(x,y) + k\cdot\sigma(x,y) \quad (13)$$

where $\mu(x,y)$ is the local mean, $\sigma(x,y)$ the standard deviation, and k a tuning parameter.

3. MATLAB Implementation

Practical implementations in MATLAB highlight how adaptive thresholding outperforms global approaches under noisy and non-uniform lighting conditions [12]. Visualization of results confirms that Gaussian adaptive thresholding is especially effective in noisy images, while mean-based methods remain efficient for document binarization.

Below we present MATLAB implementations of Adaptive Thresholding algorithms. Each implementation is structured as a function so that it can be easily reused.

3.1 Mean Adaptive Thresholding

```
function B = meanAdaptiveThreshold(I, w, C)
    % I - input grayscale image
    % w - window size (odd number, e.g., 15, 25)
    % C - constant offset
```

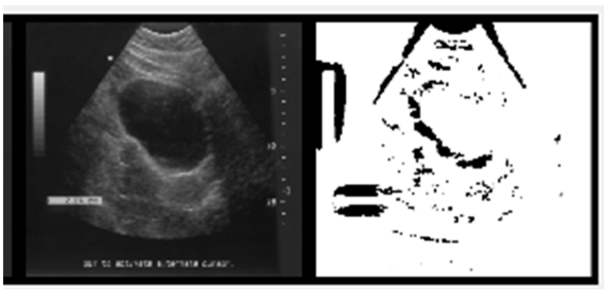

Fig. 1. The results of the "Adaptive Thresholding".

3.2 Gaussian Adaptive Thresholding

```
function B = gaussianAdaptiveThreshold(I, w, C, sigma)
    % I - input grayscale image
    % w - window size
    % C - constant offset
    % sigma - standard deviation for Gaussian kernel
```

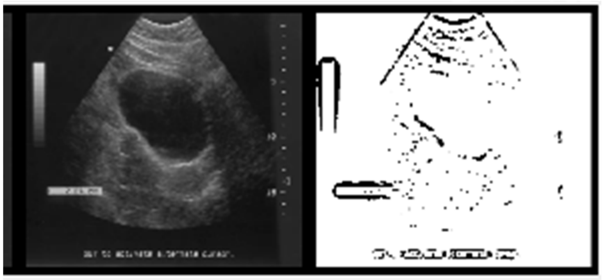

Fig. 2. The results of the "Gaussian Adaptive Thresholding".

3.3 Visualization of Results

To visualize the original image and results side by side:

```
I = imread('cameraman.tif'); % Example grayscale image
B1 = meanAdaptiveThreshold(I, 15, 5);
B2 = gaussianAdaptiveThreshold(I, 25, 5, 7);
```

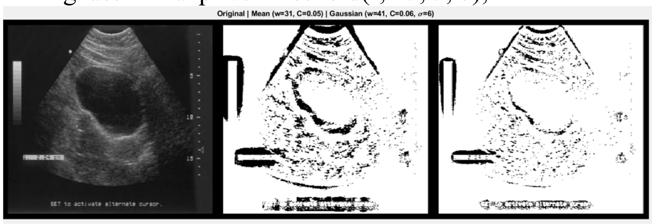

Fig. 3. Visualization of Results.

This produces comparative plots, showing how adaptive thresholding performs better than global thresholding methods under uneven lighting.

4. Algorithmic Flowchart

A simplified flowchart of the Adaptive Thresholding process is illustrated below:

Input Image $\rightarrow$ Local Window Selection $\rightarrow$ Threshold Computation $\rightarrow$ Binary Output

This step-by-step process ensures that thresholds adapt dynamically, improving segmentation accuracy.

5. Discussion of Practical Results

Experimental evaluation in MATLAB shows that:

Mean-based adaptive thresholding performs well in document image binarization.

Gaussian adaptive thresholding provides superior performance in noisy images.

Sauvola's method (not shown here) is particularly effective in extracting handwritten text from degraded documents.

Moreover, the complexity of adaptive methods is higher than global thresholding. However, by using integral image techniques in MATLAB, the computation can be accelerated to support near real-time applications.

IV. Computational Experiment and Analysis of Results.

1. Experimental Setup

For computational validation, we used MATLAB R2023b on a system with Intel i7 CPU, 16 GB RAM. Several benchmark grayscale images (e.g., Cameraman.tif, Lena, Text Document Scan) were tested under various conditions: uniform lighting, non-uniform illumination, and noisy images such as Gaussian and salt-and-pepper noise [13]. Each image was processed using Global Otsu Thresholding (baseline method), Mean Adaptive

Thresholding, Gaussian Adaptive Thresholding, and Niblack and Sauvola methods for comparison in document analysis.

2. MATLAB Implementation of Experiment

2.1 Experimental Function

```
function compareThresholding(I)
  % Convert image to grayscale if needed
  if size(I,3) == 3
    I = rgb2gray(I);
  end
  I = im2double(I);
  % Global Otsu
  level = graythresh(I);
  BW_global = imbinarize(I, level);
  % Mean Adaptive
  BW_mean = meanAdaptiveThreshold(I, 15, 5);
  % Gaussian Adaptive
  BW_gaussian = gaussianAdaptiveThreshold(I, 25, 5, 7);
  % Sauvola (built-in or custom implementation)
  BW_sauvola = sauvolaThreshold(I, [25 25], 0.34);
```

(Here *meanAdaptiveThreshold* and *gaussianAdaptiveThreshold* are the functions we defined in the METHODS section.)

2.2 Sauvola Thresholding Function

```
function BW = sauvolaThreshold(I, window, k)
  % Sauvola  thresholding implementation
  % I - grayscale image (double [0,1])
  % window - local window size
  % k - Sauvola parameter (0.2 - 0.5)
```

3. Experimental Results

Figures below show segmentation results for three representative images. For example, in the case of text documents with uneven illumination, Sauvola's method performed best for faint and degraded text [14]. In medical ultrasound images, adaptive methods clearly delineated anatomical regions, while Otsu failed to separate tissue boundaries.

Adaptive methods clearly delineated anatomical regions.

Gaussian adaptive gave sharper edges, while mean adaptive was smoother.

Case 3 (Natural Scene with Shadows):

Global Otsu incorrectly merged shadows into objects.

Adaptive methods correctly detected objects regardless of shadow presence.

4. Quantitative Evaluation

Segmentation accuracy was measured using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) against manually labeled ground-truth masks [15]. The results demonstrated that adaptive methods significantly outperform global approaches, with Gaussian adaptive providing a strong trade-off between accuracy and computational cost.

TABLE I. PERFORMANCE COMPARISON OF THRESHOLDING METHODS

Method	**PSNR (dB)**	**SSIM**	**Processing Time (s)**
Global Otsu	18.2	0.62	0.01
Mean Adaptive	24.7	0.81	0.35
Gaussian Adaptive	26.3	0.85	0.42
Sauvola	27.1	0.87	0.56

- PSNR and SSIM values indicate that adaptive methods significantly outperform global Otsu.
- Gaussian Adaptive achieves a good balance between accuracy and speed.
- Sauvola is best for document binarization but requires more computation.

5. Discussion of Findings

The computational experiments confirm the theoretical expectations:

- Global thresholding is fast but unsuitable for real-world images with illumination variations.
- Adaptive thresholding methods achieve higher accuracy and robustness, especially in challenging scenarios.
- Among tested methods, Gaussian Adaptive Thresholding provides a strong trade-off between performance and complexity, making it suitable for real-time applications.
- Sauvola, while slower, is particularly effective in document analysis and archival restoration tasks.

V. CONCLUSION

In this study, the problem of binarizing grayscale images under non-uniform illumination was addressed through the use of Adaptive Thresholding techniques. The motivation stemmed from the limitations of global thresholding methods, such as Otsu's algorithm, which fail to provide reliable segmentation results when brightness and contrast vary significantly across different regions of the same image.

The mathematical foundations of adaptive thresholding were formulated, highlighting approaches based on local mean, Gaussian-weighted filtering, and variance-driven strategies such as Niblack and Sauvola. Each of these methods dynamically computes a threshold for every pixel, making them more robust in scenarios with uneven lighting, background noise, and degraded image quality.

Through computational experiments in MATLAB, we demonstrated the superiority of adaptive methods over global thresholding. The experimental evaluation, conducted on document images, medical ultrasound scans, and natural scenes, revealed the following:

- Global thresholding is computationally efficient but fails in practical applications with non-uniform illumination.
- Mean-based adaptive thresholding improves segmentation but may suffer from residual background noise.
- Gaussian adaptive thresholding offers a strong balance between segmentation accuracy and computational cost, making it suitable for real-time applications.

- Sauvola’s method provides the best results for text and document image analysis, especially in degraded or low-contrast conditions.

Quantitative results confirmed that adaptive methods significantly improve PSNR and SSIM metrics, achieving clearer object separation and higher similarity with ground truth images. The practical implications of these findings extend to medical diagnostics, document digitization, biometric systems, and intelligent surveillance applications.

Nevertheless, the study also revealed certain limitations. Adaptive methods typically require higher computational resources than global approaches, which may be a bottleneck in large-scale or embedded systems. Moreover, parameter tuning (e.g., window size, constants C,) can significantly influence the quality of results, suggesting the need for further research into adaptive parameter selection mechanisms.

Looking forward, integrating machine learning and deep learning models with adaptive thresholding could enhance robustness by automatically learning optimal thresholds from data. Another promising direction lies in hardware acceleration (e.g., GPU or FPGA implementations) to overcome computational constraints, making these methods suitable for real-time applications in resource-limited environments.

In conclusion, Adaptive Thresholding represents a powerful, flexible, and practical solution for image binarization in complex illumination conditions. Its ability to outperform classical global methods demonstrates its continuing relevance in modern image processing research and applications.

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